

The Accuracy Illusion: How Overlapping Samples Manufacture Confidence, Not Skill, in Intraday Index Forecasting

Rachit Ranka

Department of Computer Science Engineering
Chandigarh University
Mohali, India
rankarachit5@gmail.com

Aaditya Saini

Department of Computer Science Engineering
Chandigarh University
Mohali, India
aadityasaini899@gmail.com

Abstract—Intraday forecasting systems routinely report directional hit-rates between 60 % and 85 %, and those figures move real capital. We show they are not valid estimates of skill, and measure why. A k -bar-ahead forecast issued every bar shares $k-1$ of its k outcome bars with its neighbour; when the model's calls also persist, as they do on a live system we instrument, the reported hit-rate acquires a variance far larger than its nominal sample size implies. The reading is not biased upward. Its stated precision is. A forecaster with *exactly zero skill*, simulated against real NSE index bars, posts a ≥ 71 % session once in 50 while reporting an interval $1.66\times$ too narrow, and on a two-index four-horizon panel such a reading appears every seventh session.

We give the closed form behind this, the classical design effect with clusters given by runs of identical calls, and validate corrected intervals by coverage on simulated and on live sessions: a session advertising 69 forecasts carries about twelve independent observations. Applying the correction to our own results removes our one positive directional finding, XGBoost falling from 51.3 % ($p = 0.001$) to 50.4 % ($p = 0.69$), with zero of nine model families surviving on either index. It also removes our best non-directional result, a premium-selling strategy that does not beat a recentred null over its own configuration grid. A volatility target on identical features is untouched (70.8 % to 71.3 %), so the correction distinguishes structure from artefact rather than flattening both — though that structure is volatility clustering, and its margin over a persistence baseline is not established either. Break-even accuracy derived from measured move distributions is 71.5 % at thirty minutes and 86–93 % at five, against a best achieved figure near 50 %. Code, raw results and the free-data corpus are public.

Index Terms—overlapping observations, backtest overfitting, directional forecasting, market microstructure, multiple hypothesis testing, negative results, reproducibility, index derivatives

I. INTRODUCTION

A forecasting system that reports 71 % directional accuracy is, on its face, extraordinary. Work that controls for scoring unit, transaction costs and multiple testing places realistic out-of-sample directional accuracy for liquid index intraday prediction close to the coinflip [13], [21], while

surveys of the applied literature catalogue a great many higher figures reported without those controls [32]. Yet such numbers appear regularly on trading dashboards, in vendor marketing and in applied publications, and they are frequently computed in good faith from data the system itself generated.

This paper began as an attempt to build such a system. Over fourteen months we built and ran a live multi-horizon forecasting platform for Indian index derivatives, and it reported a headline accuracy of 71 %. That number was wrong. We should say at the outset that the specific session producing it cannot be re-audited, because the portion of the forecast log holding it was lost before this analysis began; every figure we quote comes from sessions that survive, and Section VI-C works through the closest of them in full. Tracing it to source revealed no bug in the model. The defect lay in the *denominator*, in how a per-minute forecast stream is scored. Five further positive results from the same project have since been retracted, one of them in this paper.

The econometrics of overlapping observations is not new [1], [2]. Hansen and Hodrick [1] established four decades ago that sampling data more finely than the forecast interval induces serial correlation in forecast errors and invalidates naive standard errors. What that literature does not document, and what we supply here, is the *magnitude and shape* of the distortion in a now-ubiquitous practice: the every-minute multi-horizon directional dashboard. There the overlap meets a second ingredient, prediction persistence, and the product is then amplified by the multiplicity of an index $\times$ horizon panel.

A. Contributions

- 1) **A measurement of the artefact as it occurs in practice.** Overlap on its own changes nothing, since a forecaster that re-draws every bar yields an i.i.d. hit indicator. Inflation requires overlap *and* persistent predictions together. We calibrate persistence from a production system's own forecast log, then

quantify the resulting distribution, its calibration error and its panel multiplicity (Section V).

- 2) **A complete control set for a negative result.** Shuffled-label controls on every model, injected synthetic signals of known strength, nine architectures under identical purged splits, and a block-bootstrap null for the strategy sweep. Without the positive control, “we found nothing” cannot be distinguished from “our pipeline is broken.”
- 3) **A closed form and a corrected estimator.** We derive the effective sample size of an overlapping, persistent forecast stream and show it is governed by $\mathbb{E}[L^2]/\mathbb{E}[L]$ over the run lengths. An interval built on it is conservative where the conventional interval is badly anti-conservative: on a session hit-rate the naive interval covers at 75 % against a nominal 95 %, HAC and a block bootstrap at 86–87 %, and the design effect at 99.5 % (Section VI).
- 4) **The required-versus-achieved framing.** We derive break-even accuracy from measured move distributions and show that below a horizon threshold *no* accuracy is sufficient, which reframes high-accuracy claims as either artefacts or as far less valuable than they appear (Section VIII).
- 5) **A reusable free-data corpus** for NSE index derivatives: a decade of end-of-day option chains, two years of participant-wise open interest, and the observation that the exchange’s static archive is not geo-restricted while its live API is (Section III).

All experiments are reproducible from public code and committed raw results; the repository is given in the Reproducibility note at the end.

II. RELATED WORK

A. Overlapping observations

Hansen and Hodrick [1] and Richardson and Smith [2] characterise inference under overlapping returns; Newey and West [6] give the standard heteroskedasticity- and autocorrelation-consistent correction. Boudoukh, Richardson and Whitelaw [3] are the closest antecedent to this paper: they show that overlapping observations manufacture the appearance of long-horizon return predictability, and that the apparent effect largely disappears under correct inference. Our subject is the same distortion at the opposite end of the horizon scale, and applied to a different reported quantity.

That literature concerns regression coefficients and their standard errors. We apply the same logic to what practitioners actually publish, the session directional hit-rate, and measure the consequence empirically rather than asymptotically. The correction we arrive at is the classical design effect [4], [5] with clusters given by runs of identical calls; the contribution is its application to forecast streams, not the statistic itself.

B. Backtest overfitting and multiple testing

Harvey, Liu and Zhu [15] argue that the conventional $t > 2$ threshold is indefensible after decades of factor mining and propose $t > 3$. Bailey and López de Prado [16] introduce the Deflated Sharpe Ratio, and Bailey *et al.* [17] the Probability of Backtest Overfitting via combinatorially symmetric cross-validation. White’s Reality Check [12], its refinement by Hansen [14] and the Benjamini–Hochberg procedure [11] provide complementary family-wise and false-discovery-rate controls. Sullivan, Timmermann and White [13] apply the Reality Check to a large universe of technical trading rules, which is the direct precedent for our strategy sweep; Section VII-C is best read as a replication of that design on intraday index data with a modern cost model. We apply these corrections and report openly where each one fails on our data and why.

C. Machine learning for financial prediction

Fischer and Krauss [30] train LSTM networks on S&P 500 constituents; their headline performance figures are quoted gross, with transaction costs discussed separately rather than deducted. Krauss *et al.* [31] compare deep networks, gradient-boosted trees and random forests on the same universe. Gu, Kelly and Xiu [29] conduct a comprehensive machine-learning comparison for asset pricing, reporting modest out-of-sample R^2 . Zhang *et al.* [28] achieve strong accuracy on limit-order-book data, and Kolm *et al.* [27] demonstrate genuine multi-horizon alpha from order-flow imbalance. Both results rest on message-level exchange data rather than on bars, which is the distinction that matters for our purposes. Sezer *et al.* [32] survey the field.

Table I compares this work with representative prior studies along the methodological axes that determine whether a reported accuracy is interpretable. We deliberately compare on *procedure* rather than on headline numbers: our claim is not that prior results are wrong, but that several widely used reporting practices admit the artefact documented here, and that a reported accuracy cannot be assessed without knowing the scoring unit and the cost model.

D. What is predictable

Volatility clustering [22], [23] and the intraday volatility U-shape [24] are among the most robust regularities in financial time series, and they concern magnitude rather than direction. The variance risk premium, meaning the persistent gap between implied and subsequently realised volatility, is documented by Carr and Wu [25] and by Bakshi and Kapadia [26]. Our positive results fall entirely within these two categories, and we regard that as the paper’s substantive finding rather than as a consolation.

III. DATA

All data is obtainable free of charge from a retail brokerage account and public exchange archives. Table II summarises the corpus.

TABLE I

METHODOLOGICAL COMPARISON WITH REPRESENTATIVE PRIOR WORK. “SCORING UNIT” IS THE UNIT IN WHICH HIT-RATE OR RETURN IS AGGREGATED; PER-BAR SCORING OF A MULTI-BAR HORIZON IS THE CONDITION UNDER WHICH THE ARTEFACT OF SECTION V APPEARS. MT: MULTIPLE-TESTING CORRECTION. NC: NULL CONTROL (SHUFFLED LABELS OR EQUIVALENT). C: TRANSACTION COSTS MODELLED. R: CODE AND RAW RESULTS PUBLIC.

Study	Market	Data granularity	Scoring unit	MT	NC	C	R
Fischer & Krauss [30]	S&P 500	Daily bars	Per day, non-overlapping	—	—	pre-cost	—
Krauss <i>et al.</i> [31]	S&P 500	Daily bars	Per day, non-overlapping	—	—	pre-cost	—
Gu <i>et al.</i> [29]	US equities	Monthly	Per month	partial	—	partial	partial
Zhang <i>et al.</i> [28]	LSE	Message-level LOB	Per event	—	—	—	partial
Kolm <i>et al.</i> [27]	Nasdaq	Message-level LOB	Per horizon	—	—	discussed	—
This work	NSE index	5-min bars, EOD chains	Per event (k-bar)	✓	✓	✓	✓

MT here means Bonferroni and Benjamini–Hochberg, which applied cleanly, plus the Deflated Sharpe Ratio and PBO/CSCV, which we ran and found inapplicable and degenerate respectively (Section VII-C). We omit a row for “typical applied studies”: characterising that population would require citing specific papers as negative examples, and the argument does not need it.

TABLE II

FREE-DATA CORPUS. ALL COLLECTORS ARE PUBLIC.

Asset	Period	Size
BANKNIFTY 5-min bars	2025-02 – 2026-07	26 265
NIFTY 50 5-min bars	2025-02 – 2026-07	26 267
FINNIFTY / MIDCPNIFTY / SENSEX	2025 – 2026	9.6k–14k
Option chain EOD + OI	2024-07 – 2026-07	1 382 738
Long-history option chain	2016-05 – 2026-05	2 181 d
Participant-wise OI	2024-07 – 2026-07	515 d
Index futures (real volume)	2024-01 – 2026-07	639 d

Three observations about this corpus are of independent practical value.

Index spot carries no volume. The cash index prints zero volume, so a feature computed as a “VWAP” on index spot is an unweighted expanding mean of typical price. This sat in our own feature set for weeks. Real volume exists only on the futures contract.

The static exchange archive is not geo-restricted. The *live* option-chain API returns HTTP 403/404 from outside India; the static end-of-day archive downloads normally. That distinction turned an assumed hard constraint into a ten-year dataset.

TOTP authentication is location-independent. Brokers authenticating by locally computed one-time passwords permit collection from any jurisdiction, whereas SMS-based flows do not. That is a deciding constraint on reproducibility and it is documented nowhere prominent.

IV. METHODOLOGY

Every accuracy reported in this paper passes through one shared harness, so that differences between models are differences between models.

Purged walk-forward. Expanding-window splits, with the k bars whose labels overlap the test window removed from the end of each training set. Label overlap is the most common leakage channel in this setting [18].

Exact intervals. Accuracies are reported with Clopper–Pearson [39] 95% confidence intervals and the exact binomial p -value against 0.5. We additionally report the Pesaran–Timmermann statistic [10], which is the

appropriate null for directional accuracy: it compares the realised hit-rate against what independence between forecast and outcome would produce given their own marginal rates, and therefore catches a model that merely tracks drift.

Shuffled-label controls. Every model is additionally trained and evaluated with randomly permuted labels. If real and shuffled scores coincide, the pipeline is reading noise. That is a stronger statement than any train/test gap, since it does not require the model to be well specified.

Positive controls. A synthetic feature agreeing with the label on a known fraction s of rows is injected, making the recoverable accuracy approximately $0.5 + s/2$. A pipeline that fails to recover it cannot be trusted to report a null.

Per-event scoring and costs. Strategy returns are evaluated on non-overlapping k -bar holds with explicit round-trip costs, and costs are treated as part of the hypothesis rather than as a subsequent adjustment.

Sample sizes. Three counts appear for the same experiment and are distinguished here once. BANKNIFTY has 26 265 raw 5-minute bars; 24 080 survive feature warm-up and the removal of bars whose k -ahead label would cross a session boundary; and 16 856 of those fall in a test fold under the expanding walk-forward, which is the n every accuracy is computed on.

Units. One conversion is used throughout and is stated here once to avoid ambiguity later. All analysis is on 5-minute bars, so the 30-minute horizon is $k = 6$ bars and a session of 09:15–15:30 yields $n = 69$ scored forecasts. The production system emits a forecast every *minute*, so its run lengths are measured in minutes and divided by five to obtain bars. Pooled across both indices the mean run is 13.5 minutes, or 3.0 bars after rounding, and the longest is 207 minutes, or 41 bars. Per index the means are 15.2 minutes (BANKNIFTY) and 12.2 (NIFTY 50), with longest runs of 135 and 207 respectively; simulations use the pooled distribution and quote the pooled figures. Because the live stream is five times denser than our simulation while carrying the same information, the

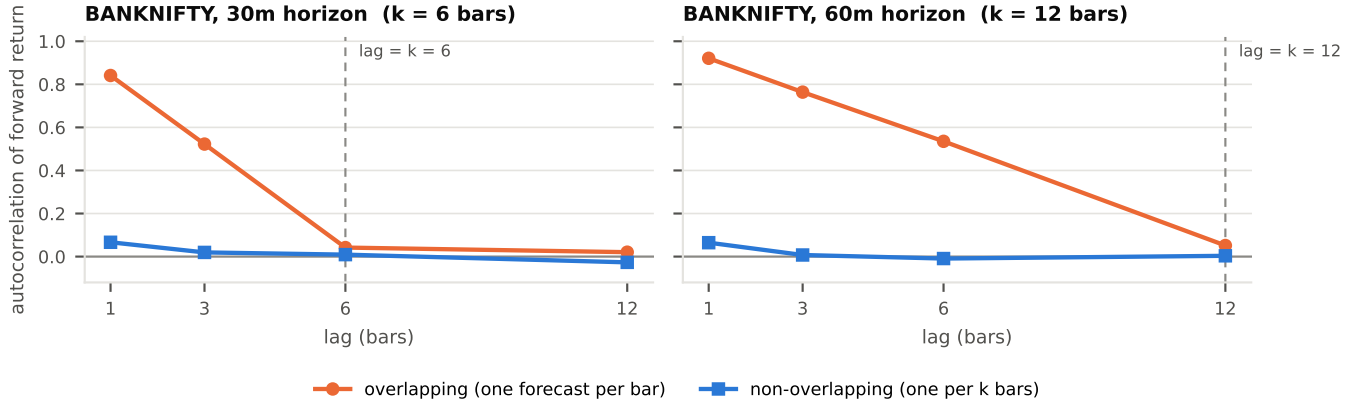


Fig. 1. Autocorrelation of the forward return, overlapping versus non-overlapping, at two horizons. In both panels the overlapping series decays to zero at exactly lag k , which is the signature of arithmetic overlap rather than of serial structure.

nominal-to-effective ratio it reports is worse than the one we measure, so our figures are conservative with respect to the artefact they describe.

Features are the fifteen columns used by the production system: one-bar return, RSI, EMA distance, MACD histogram, Bollinger width and %b, ATR%, Parkinson volatility, VWAP distance, time-of-day, day-of-week, regime volatility, Hurst exponent, and distance from the previous session’s high and low. All are transforms of a single price series, a point we return to in Section IX.

V. THE MECHANISM

A. Overlap is arithmetic, not information

Fig. 1 shows the autocorrelation of the k -bar forward return measured two ways. Computed on overlapping windows it appears strongly autocorrelated, at 0.841 for lag 1, and decays to approximately zero at *exactly* lag k . Sampled so that no two observations share a bar, the autocorrelation is 0.067 at lag 1 and indistinguishable from zero thereafter. The pattern reproduces independently at $k = 6$ (30 minutes) and $k = 12$ (60 minutes), with the decay landing on lag 6 and lag 12 respectively. The correlation is a property of the sampling scheme, not of the market.

B. Persistence is the second, necessary ingredient

Overlap alone is insufficient. If predictions are drawn independently each bar, the hit indicator $\mathbf{1}\{\hat{y}_t = y_t\}$ is i.i.d. Bernoulli regardless of how autocorrelated y_t is, and no inflation occurs. Our first simulation made precisely this error and correctly found nothing.

Real models do not re-decide every bar: their features move slowly, so they repeat a call for long stretches. Measured on the production system’s own forecast log at the 30-minute horizon, BANKNIFTY calls persist for a mean of 15.2 minutes (99 runs across 1 500 forecasts, longest 135 minutes) and NIFTY 50 for 12.2 minutes (123

TABLE III
A FORECASTER WITH EXACTLY ZERO SKILL, 30-MINUTE HORIZON, SCORED PER BAR VERSUS PER EVENT. BANKNIFTY; NIFTY 50 IS WITHIN 0.3 POINTS ON EVERY ROW.

	Per bar	Per k bars
Reported sample size per session	69	12
Stated binomial 95 % CI	± 11.8 pts	± 28.3 pts
Realised spread across sessions	± 19.6 pts	± 28.0 pts
Understatement factor	1.66$\times$	0.99$\times$
$P(\text{session} \geq 60 \%)$	15.7 %	19.2 %
$P(\text{session} \geq 71 \%)$	2.0 %	7.1 %
$P(\text{session} \geq 85 \%)$	≈ 0.01 %	0.30 %

runs, longest 207 minutes). A model re-drawing each bar would show a mean run of one.

When both the call and the outcome persist, a single trending stretch is recorded as dozens of hits.

C. What a zero-skill forecaster reports

We simulate a forecaster with no skill whatsoever: a fair coin, held for a run length drawn from the empirical distribution above. It is evaluated against real BANKNIFTY and NIFTY 50 bars and scored the way a live dashboard scores it. Results over approximately 70 000 simulated sessions per index appear in Table III and Fig. 2.

Table III makes a point that is easy to state backwards, so we state it carefully. Per-bar scoring does *not* produce extreme session readings more often than honest scoring. It produces them *less* often: a ≥ 71 % session appears on 2.0 % of per-bar sessions against 7.1 % when scored per event. What per-bar scoring does is report those readings with an interval 1.66 $\times$ too narrow, because the quoted n of 69 is a fiction.

So overlap does not manufacture high hit-rates. It manufactures unwarranted *confidence* in them. A figure that should be read as “somewhere between incompetence and mastery” is instead read as a precise 71 %, and it is that

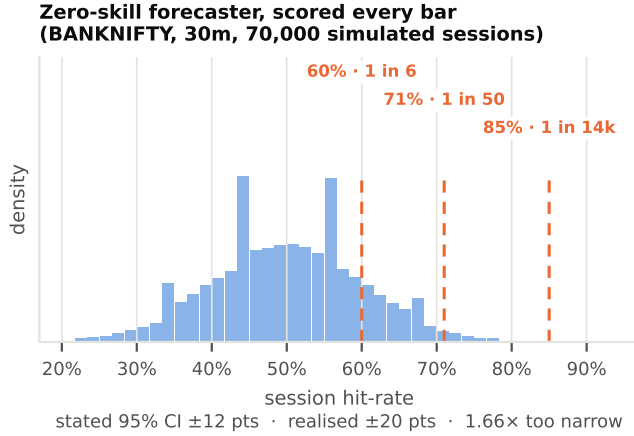


Fig. 2. Measured session hit-rate distribution of a zero-skill forecaster scored every bar. The multimodality is not noise: with roughly a dozen independent decisions per session the hit-rate lands on near-discrete values, which is the small effective sample made visible.

TABLE IV

THE SAME ZERO-SKILL FORECASTER AT BOTH GRANULARITIES, 30-MINUTE HORIZON. PER-MINUTE OUTCOME SERIES ARE TAKEN FROM THE PRODUCTION LOG; RUN LENGTHS ARE THE POOLED PER-MINUTE RUNS (MEAN 13.5 MIN, LONGEST 207). FORECASTS PER SESSION ARE AVERAGED OVER THE LOGGED SESSIONS, WHICH RANGE FROM 245 TO 336.

	per 5-min bar	per minute
Forecasts per session	69	314
Design-effect n_{eff}	12.4	16.5
Stated s.d.	6.02 %	2.82 %
Realised s.d.	10.00 %	8.76 %
Understatement factor	1.66×	3.11×
$P(\text{session} \geq 71 \%)$	2.0 %	0.6–1.3 %

false precision, not the figure itself, that misleads. The calibration column is therefore the diagnostic throughout this paper, and the frequency columns are context.

D. At the granularity it is actually reported

Everything above runs on 5-minute bars, where a 30-minute horizon is $k = 6$ and a session yields 69 forecasts. The production system does not work that way: it emits a forecast every minute, so the horizon is $k = 30$ and a session yields about 314 scored forecasts on average, and over 330 on a full session; the headline figures this paper exists to explain were counted at that density. Explaining a per-minute figure with a per-bar model would be a gap, so we close it (Table IV).

The outcome series here is not simulated. It is taken from the production log, which recorded the direction called each minute and the move that followed, so the truth carries its real per-minute autocorrelation rather than an assumed one. Only the forecaster is synthetic, and it has zero skill by construction.

Two things follow, and they point the same way as Section V-C. The stated interval is *worse* calibrated at

the real granularity, $3.11\times$ too narrow against $1.66\times$: the nominal count rises roughly fivefold while the information does not. But extreme sessions become slightly *rarer*, not commoner. Per-minute scoring does not manufacture more 71 % readings; it manufactures more confidence in the ones it produces. Every 5-minute-bar figure elsewhere in this paper therefore understates the artefact rather than exaggerating it.

E. Panel multiplicity

A dashboard does not monitor one number. The system studied here displays two indices at four horizons, so eight hit-rates every session. The two indices are strongly correlated intraday and the four horizons nest, so an independence calculation would be an upper bound of unknown tightness. We therefore simulate the whole panel jointly on the same sessions, each horizon carrying its own k and its own forecast stream, and count the joint event directly:

the per-series rate is 2.04 %, an independence calculation gives 15.2 %, and the measured panel gives **15.1 %**. The approximation is adequate here, which is worth stating: although the outcome series are correlated, a zero-skill forecaster’s hits are close to independent across them. Either way the figure is about one session in seven, so a headline 71 % arrives within the first weeks of live operation with no edge whatsoever.

Our own headline figure was exactly this: one index, one horizon, one trending Friday, read as skill. That session cannot be audited, since the portion of the log holding it was lost before this analysis began, and we will not quote figures we cannot reproduce in a paper arguing against the practice. The surviving log holds a close analogue which Section VI-C takes apart in full: 69.6 % over 335 per-minute NIFTY 50 forecasts, whose naive interval excludes 50 % comfortably and whose corrected interval barely does.

VI. A CORRECTED ESTIMATOR

Diagnosing a biased statistic is only half a contribution. This section supplies the closed form behind the distortion and an estimator that removes it, then validates the estimator by coverage rather than by argument.

A. Effective sample size

Let $h_t = \mathbf{1}\{\hat{y}_t = y_t\}$ denote the hit indicator of a forecaster issuing a k -bar-ahead call every bar, and let $\hat{p} = n^{-1} \sum_t h_t$ be the reported hit-rate. Practitioners quote $\text{Var}(\hat{p}) = p(1-p)/n$, which holds only for independent h_t . In general,

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \left[1 + 2 \sum_{j=1}^{n-1} \left(1 - \frac{j}{n} \right) \rho_j \right], \quad (1)$$

where $\rho_j = \text{Corr}(h_t, h_{t+j})$. The bracketed term is the variance inflation factor, and dividing n by it gives the effective sample size n_{eff} .

TABLE V

COVERAGE OF A NOMINAL 95 % INTERVAL ON A SESSION HIT-RATE, AGAINST A FORECASTER OF KNOWN SKILL WHOSE CALLS PERSIST PER THE MEASURED RUN-LENGTH DISTRIBUTION. BANKNIFTY, 30-MINUTE HORIZON, NOMINAL $n = 69$; NIFTY 50 AGREES WITHIN 2 POINTS THROUGHOUT. TRUE SKILL IS MEASURED FROM THE GENERATIVE PROCESS RATHER THAN ASSUMED. HAC-L IS THE BARTLETT KERNEL WITH ITS BANDWIDTH TAKEN FROM THE RUN STRUCTURE INSTEAD OF FROM n ; CALIBRATED DIVIDES THE DESIGN EFFECT BY TWO.

True skill	Naive	HAC	HAC-L	Boot	Design	Calib.	n_{eff}
50 %	75.5	87.0	88.4	86.6	99.5	96.0	12.4
55 %	74.8	86.6	88.2	86.4	99.6	96.1	12.0
59 %	74.0	85.6	87.0	85.2	99.6	95.8	11.2

All figures are coverage in per cent of a nominal 95 %. n_{eff} is the raw design-effect estimate; the calibrated variant uses twice this.

Two structures drive ρ_j , and the argument of Section V is that both are required. The outcome y_t is autocorrelated out to lag k because consecutive windows share bars, and the prediction $\hat{y}_t$ persists in runs. If predictions are redrawn each bar, h_t is i.i.d. whatever y_t does, every ρ_j vanishes, and $n_{\text{eff}} = n$.

Because the dependence here is block-shaped, (1) admits a direct form. If the sequence decomposes into runs of length L_i within which h_t is constant, then conditioning on the lengths gives $\text{Var}(\hat{p}) = p(1-p) \sum_i L_i^2/n^2$, so

$$n_{\text{eff}} = \frac{n}{\sum_i L_i^2/n} \xrightarrow{n \rightarrow \infty} \frac{n \mathbb{E}[L]}{\mathbb{E}[L^2]}. \quad (2)$$

Equation (2) is the classical design effect [4], [5] with clusters given by the runs; what is new here is only the observation that a forecast stream has exactly this structure and that nobody reports it.

The inflation is governed by $\mathbb{E}[L^2]$, not $\mathbb{E}[L]$. A run-length distribution with a heavy tail inflates the variance far beyond what its mean suggests, and that is precisely the regime a real forecaster occupies: on our production log the pooled mean run is 13.5 minutes while the longest is 207.

B. Validation by coverage

An interval is judged by whether it covers. We generate a forecaster whose true skill is fixed by construction, with calls persisting according to the measured run-length distribution, and ask how often a nominal 95 % interval contains that known truth. Table V reports four estimators.

Both HAC and the bootstrap improve substantially on the naive interval and both still under-cover by around eight points. The obvious diagnosis is bandwidth: the automatic Bartlett rule $4(n/100)^{2/9}$ evaluates to 3.7 bars at $n = 69$, almost exactly the pooled *mean* run length of 3.0 bars, whereas the inflation is driven by $\mathbb{E}[L^2]$ and the pooled tail reaches 41 bars. A kernel truncated near the mean sums almost none of the covariance the long runs contribute.

TABLE VI

FOUR INTERVALS ON THE TWO BEST SESSIONS IN THE PRODUCTION LOG, 30-MINUTE HORIZON. THE FINAL ROW IS THE RANGE OF HIT-RATES OBTAINED FROM THE $k = 6$ POSSIBLE PER-EVENT SUBSAMPLES OF THE SAME SESSION.

Estimator	95 % interval	Width	n_{eff}
<i>BANKNIFTY, 3 Aug: 336 forecasts, 57.1 %</i>			
Naive binomial at n	[51.9, 62.4]	10.6	336
HAC (Bartlett)	[41.8, 72.4]	30.6	40
Block bootstrap	[41.7, 69.3]	27.7	—
Run-structure	[34.4, 79.9]	45.5	18
Per-event, 6 subsamples	53.6 to 62.5	8.9	56
<i>NIFTY 50, 3 Aug: 335 forecasts, 69.6 %</i>			
Naive binomial at n	[64.6, 74.5]	9.9	335
HAC (Bartlett)	[54.8, 84.3]	29.6	37
Block bootstrap	[57.9, 79.4]	21.5	—
Run-structure	[50.0, 89.1]	39.1	21
Per-event, 6 subsamples	66.1 to 73.2	7.1	55

We tested that diagnosis rather than asserting it. Giving the Bartlett kernel a bandwidth taken from the run structure (HAC-L in Table V) recovers between one and two points of coverage and no more. Bandwidth was part of the problem and not the whole of it, and we report the repair even though it mostly fails.

The design effect (2) removes the failure mode entirely and over-corrects in the other direction, reaching 99.5 %. The reason is structural: treating runs as independent clusters ignores that consecutive runs are negatively dependent, since a run of hits ends precisely when a miss begins. That over-statement is close to a constant factor of two, and it is visible from two directions at once. The realised spread in Table III implies a true effective sample near 25 for a 69-forecast session where (2) estimates 12, and coverage sits at 99.5 % rather than 95 %.

Dividing the inflation by that factor gives the *calibrated* estimator, which covers at 95.8–96.1 % across all three skill levels and both indices. The constant is fixed in advance rather than tuned per sample, and it is judged only by whether coverage lands at nominal, which it does. We therefore recommend the calibrated interval and report the raw design effect as the conservative bound it is.

C. A real session

Everything above is simulated. Table VI applies the four estimators to sessions taken directly from the production log: the forecast stream the dashboard actually wrote, scored against the moves that actually followed.

The BANKNIFTY session reports 57.1 % with a naive interval that excludes 50 %, and would be read as evidence of skill. At $n_{\text{eff}} = 18$ the interval runs from 34 % to 80 % and supports nothing.

The NIFTY 50 session at 69.6 % is the nearest surviving analogue to the figure that started this project, and it is worth following to its end because the answer is not the convenient one. Its corrected interval is [50.0, 89.1], which excludes 50 % — but only just, and Section VI-E shows that a single session's n_{eff} is itself uncertain by roughly $0.65\times$ to $1.67\times$. Propagating that uncertainty:

n_{eff} assumption	interval	verdict
As estimated (21.3)	[50.0, 89.1]	excludes 50 %
5th percentile (13.8)	[45.3, 93.8]	includes 50 %
95th percentile (35.6)	[54.4, 84.7]	excludes 50 %

The lower bound of 50.0 is smaller than the estimator’s own noise, so the session does not support a claim of skill. An earlier draft of this paper reported that it did, on the grounds that the arithmetic said so. That was the same mistake the paper is about, committed on the last positive number left in it.

D. Why not simply score per event

Per-event scoring is unbiased, requires no estimator, and is correctly calibrated at $0.99\times$ (Table III). It is what we use for every backtest in this paper. It is nonetheless not sufficient for a live stream, and the last row of Table VI shows why.

A dashboard must print a number every bar. It cannot wait k bars before reporting, and it cannot discard $k - 1$ of every k forecasts it has already made and already shown.

Subsampling discards information. Keeping one observation in six throws away 83 % of the data, so the per-event interval is wider than the evidence requires. The corrected estimator uses the whole stream.

The choice of subsample is arbitrary, and it moves the answer. There are k valid per-event subsamples of any session. Across the six offsets, the same BANKNIFTY session reports between 53.6 % and 62.5 % and the same NIFTY 50 session between 66.1 % and 73.2 %, a spread of 7 to 12 points. An analyst free to choose the offset is free to choose the result, which is the problem this paper is about reappearing in the remedy.

E. How reliably is n_{eff} itself estimated?

Equation (2) depends on $\mathbb{E}[L^2]$, and a second moment driven by the tail is exactly what a short sample estimates worst. A single session contains only a few dozen runs. An estimator nobody can sanity-check is not practical, so we measure the error directly: the reference is the inflation computed on hit sequences pooled across many sessions, and the comparison is what each single session reports on its own.

	pooled	median	5th	95th
BANKNIFTY	11.0	12.1	7.1	18.4
NIFTY 50	10.9	12.0	7.0	18.2

A single session’s n_{eff} lands between $0.65\times$ and $1.67\times$ the pooled value nine times in ten, with a slight upward bias. That is a real limitation and it carries a practical instruction: estimate n_{eff} from a run of sessions rather than from the one being reported. Estimated from a single session it is trustworthy as an order of magnitude, single or low double digits against a nominal count in the hundreds, and not as a precise divisor.

We note that this uncertainty is small next to the quantity it is correcting. Mis-estimating n_{eff} by a factor of

TABLE VII

MODEL-INDEPENDENT CEILING TEST AT THE 30-MINUTE HORIZON, REPLICATED ACROSS FIVE INDICES. TRAINING ACCURACY IS 100.0 % ON EVERY ROW. Δ IS THE ABSOLUTE GAP BETWEEN THE REAL AND SHUFFLED COLUMNS, COMPUTED BEFORE ROUNDING. THE POSITIVE CONTROLS HAVE RECOVERABLE ACCURACIES OF APPROXIMATELY 55 % AND 65 % BY CONSTRUCTION.

Index	n	Real	Shuf.	Δ	Maj.	$s=.10$	$s=.30$
BANKNIFTY	16 856	50.5	50.2	0.3	49.3	52.4	63.5
NIFTY 50	16 853	50.0	50.2	0.3	49.8	52.0	63.7
FINNIFTY	8 988	50.2	50.5	0.3	49.4	52.4	62.5
MIDCPNIFTY	8 309	50.2	50.8	0.7	49.7	53.3	62.0
SENSEX	6 144	49.7	50.1	0.5	49.0	52.5	62.1

No real-label result is significant against 0.5 ($p = 0.22$ to 0.91); every positive control is, at $p < 0.001$.

1.7 widens or narrows an interval by $\sqrt{1.7} \approx 1.3\times$; using n instead of n_{eff} narrows it by $3.1\times$ at the granularity the system actually reports.

F. What the corrected number says

The practical content is n_{eff} itself. A session that reports 69 forecasts carries roughly a *dozen* independent observations, and an honest interval on its hit-rate is ± 12 to ± 28 points wide depending on the estimator. Even the narrowest of those spans a range within which incompetence and competence are indistinguishable, which is the correct description of what one session of a per-minute forecast stream can establish.

The correction changes no point estimate. Only the interval moves. What it buys is an interval that means what it claims, and the discipline of reporting n_{eff} alongside n whenever forecasts overlap.

VII. PREDICTABILITY UNDER CORRECT SCORING

A. Is there anything to predict?

Short-horizon index direction is where efficient-markets reasoning [19] and the random-walk tests that followed it [20] both expect to find nothing, so what is useful here is measurement rather than surprise.

We first ask the question in a form independent of any model. An unconstrained random forest was fitted until it memorised the training data, on five indices spanning large-cap, banking, financial-services, midcap and a second exchange (Table VII).

The pattern is identical on all five. Each model memorises its training data completely and generalises at chance, and permuting the labels moves out-of-sample accuracy by between 0.3 and 0.7 points. No real-label result is distinguishable from a coinflip. Both positive controls are recovered on every index at $p < 0.001$, which establishes that the pipeline detects signal whenever signal is present and that the null is therefore a property of the data.

Replication across five instruments, two exchanges and four capitalisation segments is what separates this from a single-market curiosity.

TABLE VIII

OUT-OF-SAMPLE DIRECTIONAL ACCURACY AT THE 30-MINUTE HORIZON, SCORED THREE WAYS ON IDENTICAL PURGED WALK-FORWARD SPLITS. *PER-BAR* IS WHAT A DASHBOARD REPORTS: ONE FORECAST PER BAR OVER A 6-BAR HORIZON, WITH THE EXACT BINOMIAL INTERVAL ITS NOMINAL $n = 16856$ IMPLIES. AT n_{EFF} IS THE SAME POINT ESTIMATE WITH THE INTERVAL WIDENED BY THE DESIGN EFFECT OF (2). *PER-EVENT* RE-SCORES ONE OBSERVATION EVERY 6 BARS, SO NO TWO OBSERVATIONS SHARE A BAR. SHUFFLED-LABEL CONTROLS ARE *PER-BAR*.

Model	BANKNIFTY					NIFTY 50				
	per-bar	p	CI at n_{eff}	per-event	p	per-bar	p	CI at n_{eff}	per-event	p
Majority-class baseline	49.3	0.061	—	—	—	49.8	0.655	—	—	—
Logistic regression	50.6	0.095	[48.6, 52.7]	50.1	0.895	49.3	0.082	[47.3, 51.3]	49.1	0.355
Gradient boosting	50.7	0.059	[48.9, 52.6]	50.3	0.777	49.7	0.405	[47.8, 51.6]	48.9	0.250
LightGBM	51.1	0.006	[49.2, 52.9]	50.2	0.865	49.8	0.666	[48.0, 51.7]	50.3	0.748
XGBoost	51.3	0.001	[49.5, 53.2]	50.4	0.692	50.1	0.817	[48.2, 52.0]	49.8	0.836
k -nearest neighbours	50.5	0.209	[48.7, 52.3]	49.6	0.720	50.1	0.805	[48.3, 51.9]	50.9	0.355
LSTM	50.4	0.251	[48.4, 52.5]	51.1	0.235	49.4	0.127	[47.3, 51.6]	49.4	0.509
Transformer	50.8	0.028	[48.9, 52.8]	50.9	0.355	48.3	0.000	[46.3, 50.4]	48.4	0.101
Temporal Fusion Transformer	49.4	0.101	[47.2, 51.5]	49.0	0.299	50.1	0.723	[48.0, 52.2]	51.2	0.193
Significant at $p < 0.05$	3/8		0/8	0/8		1/8		0/8	0/8	

$n_{\text{eff}} \approx 2100\text{--}3000$ against a nominal 16856; per-event $n = 2810$. Shuffled controls span 49.5–50.3% throughout. Pesaran–Timmermann p -values track the per-bar column to within 0.042 on every row (e.g. XGBoost 0.001, TFT 0.104) and are omitted for space; full values in the results file.

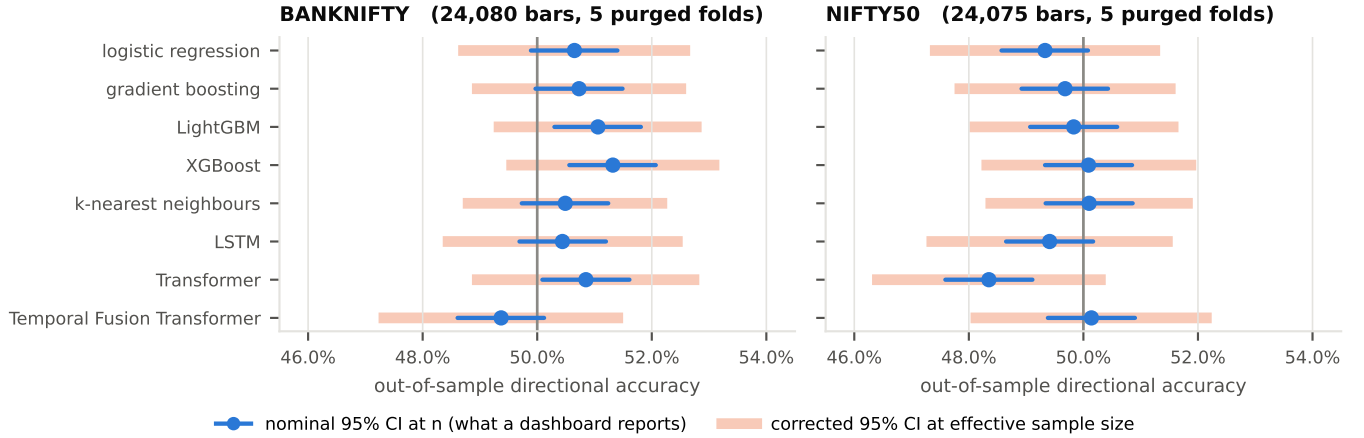


Fig. 3. Nine model families at the 30-minute horizon. The narrow blue interval is the exact binomial interval a dashboard would report at the nominal $n = 16856$; the wide orange band is the same estimate at its effective sample size. Every corrected interval contains 50%. The spread between architectures is smaller than the width of a single corrected interval, which is the sense in which the choice of architecture does not matter here.

B. Nine model families

Table VIII reports nine families on identical splits, features, seeds and controls, scored three ways: logistic regression, gradient boosting, LightGBM [37], XGBoost [36], an unconstrained forest of the Breiman type [38] (also the ceiling test’s over-capacity model), k -nearest neighbours, an LSTM [34], a Transformer [35], and a compact Temporal Fusion Transformer [33]. Fig. 3 plots the nominal and corrected intervals together.

Scored the way a dashboard scores, three of eight BANKNIFTY models clear $p < 0.05$ and XGBoost reaches 51.3% at $p = 0.001$. That was, in an earlier draft of this paper, our one surviving positive directional result, and we described it as genuine and Bonferroni-robust.

It does not survive our own correction. Widening the interval to n_{eff} puts 50% inside every interval in the table. Re-scoring one observation per 6 bars, so that no two

observations share an outcome bar, moves XGBoost from 51.3% ($p = 0.001$) to 50.4% ($p = 0.692$). Across both indices, the count of models significant at $p < 0.05$ falls from four to **zero** under either correction.

The same holds at every horizon we ran. Across four horizons, two indices and eight model families — 48 tests in all — **16 clear $p < 0.05$ scored per bar and none clears it at an effective sample size or per event** (Table IX). The count of apparent discoveries rises with horizon, from none at 15 minutes on BANKNIFTY to five at 60 minutes on both indices, which is what one expects if the artefact scales with k ; the count of surviving discoveries stays at zero throughout.

We report this prominently because it is the paper’s own thesis applied to the paper’s own result, and because a correction that spares the author’s favourite finding would be worth very little. The directional null is not a near-null

TABLE IX

MODEL FAMILIES SIGNIFICANT AT $p < 0.05$, BY HORIZON AND SCORING RULE. EIGHT FAMILIES PER CELL, EXCLUDING THE MAJORITY-CLASS BASELINE. THE 5-MINUTE HORIZON IS $k = 1$, WHERE THERE IS NO OVERLAP TO CORRECT AND PER-BAR AND PER-EVENT SCORING COINCIDE BY CONSTRUCTION.

Horizon, index	Per bar	At n_{eff}	Per event
15 m BANKNIFTY	0	0	0
15 m NIFTY 50	2	0	0
30 m BANKNIFTY	3	0	0
30 m NIFTY 50	1	0	0
60 m BANKNIFTY	5	0	0
60 m NIFTY 50	5	0	0
Total (48 tests)	16	0	0

TABLE X

STRATEGY SWEEP WITH MULTIPLICITY CONTROL, BANKNIFTY, 86 TRIALS, NON-OVERLAPPING 30-MINUTE HOLDS AT 4 BP COST.

Quantity	Value
Strategies evaluated	86
Positive Sharpe at all	2
Significantly profitable (one-sided)	0
Surviving Bonferroni	0
Surviving Benjamini–Hochberg, FDR 5 %	0
Positive <i>and</i> stable in both halves	1
Best observed Sharpe	0.21
Bootstrap null (500 resamples), median	0.44
Bootstrap null, best-of-86, 95th pct.	1.31
$P(\text{null} \geq \text{observed})$	0.658

with one exception. It is a null.

Table VIII also contains its own diagnostic. The NIFTY 50 Transformer sits at 48.3 % with $p < 0.001$, which at the nominal n is 4.4 standard errors *below* chance. Under a correctly specified null the largest $|z|$ across eighteen tests should land near 2.6–2.9; observing 4.4 is not a bug in that model but the variance inflation of Section V surfacing inside our own results table. At n_{eff} its interval is [46.3, 50.4] and the anomaly disappears.

C. Eighty-six strategies and the statistics they require

Reporting the best of a large sweep is selection, not discovery. We evaluated 86 parameterised rules, covering momentum and mean reversion at ten lookbacks, six moving average crossover pairs, RSI, Bollinger, breakout and fade variants, MACD, gap, volatility regime, time-of-day, day-of-week and streak rules, on non-overlapping 30-minute holds with 4 bp round-trip cost (Table X).

The best strategy in the sweep is *worse* than what selection alone typically manufactures. Two methodological notes:

One-sided tests. An earlier version of this analysis used a two-sided t -test and reported “42 of 86 significant.” Those were strategies that reliably *lose*. The hypothesis is profitability, so the test is one-sided; reliable losers are the cost of trading noise, not discoveries.

Where the standard corrections fail. The Deflated Sharpe Ratio [16] estimates its null from the cross-

TABLE XI

BREAK-EVEN ACCURACY FROM MEASURED MOVE DISTRIBUTIONS AT 4 BP ROUND-TRIP COST. p_{MED}^* ASSUMES A SYMMETRIC BRACKET AT THE MEDIAN MOVE; p_{MEAN}^* IS THE REQUIREMENT FOR A HOLD-TO-HORIZON RULE WITH NO BRACKET, WHICH DEPENDS ON THE MEAN ABSOLUTE MOVE. “ACHIEVED” IS THE BEST *PER-EVENT* ACCURACY OF THE NINE MODEL FAMILIES AT THAT HORIZON.

Horizon	Median	Mean	p_{med}^*	p_{mean}^*	Achieved
<i>BANKNIFTY</i>					
5 min	22.1	30.8	>100	86.4	—
15 min	37.7	52.9	79.8	71.3	51.3
30 min	52.3	73.8	71.5	65.3	51.1
60 min	72.9	103.8	65.4	60.8	52.2
<i>NIFTY 50</i>					
5 min	8.1	11.5	>100	93.0	—
15 min	14.1	19.8	84.9	74.9	50.5
30 min	19.7	27.8	75.1	67.8	51.2
60 min	27.7	38.5	67.7	62.7	50.1

Moves in index points; accuracies in per cent. Move distributions are right-skewed, with mean/median ≈ 1.4 at every horizon.

sectional variance of trial Sharpes, assuming a common-variance normal sample. Our library violates this badly: strategies differ by an order of magnitude in turnover, so their Sharpe dispersion is dominated by deterministic cost differences rather than by sampling noise, and the formula returns an implausible null benchmark of 7.77. We therefore replaced it with a circular block bootstrap [8], preferred over the moving-block [7] and stationary [9] variants because fixed-length wrapped blocks weight every observation equally. It resamples forward returns in blocks while holding each strategy’s positions and turnover fixed, destroying the signal–outcome relationship and nothing else. Similarly, PBO via CSCV [17] is *degenerate* here. With nothing significantly profitable in-sample there is no selection to overfit, so we report it as such rather than as a result. Reporting a degenerate statistic as a finding would be an instance of the error this paper is about.

VIII. ECONOMIC SIGNIFICANCE

Directional accuracy is the field’s headline metric and is, alone, close to meaningless. Whether a given accuracy earns money depends on the size of the move relative to the cost of capturing it.

For a symmetric bracket with target and stop both at the typical move m and round-trip cost c , expectancy per trade is

$$\mathbb{E} = p(m - c) - (1 - p)(m + c) = m(2p - 1) - c,$$

so break-even requires

$$p^* = \frac{1}{2} + \frac{c}{2m}. \quad (3)$$

The ratio c/m is therefore the entire problem. Table XI and Fig. 4 evaluate (3) on measured move distributions.

Two consequences follow immediately.

At the shortest horizon the requirement is out of reach, and under one model it is unreachable in principle. The median five-minute move is smaller than the round trip, so under a symmetric bracket p^* exceeds

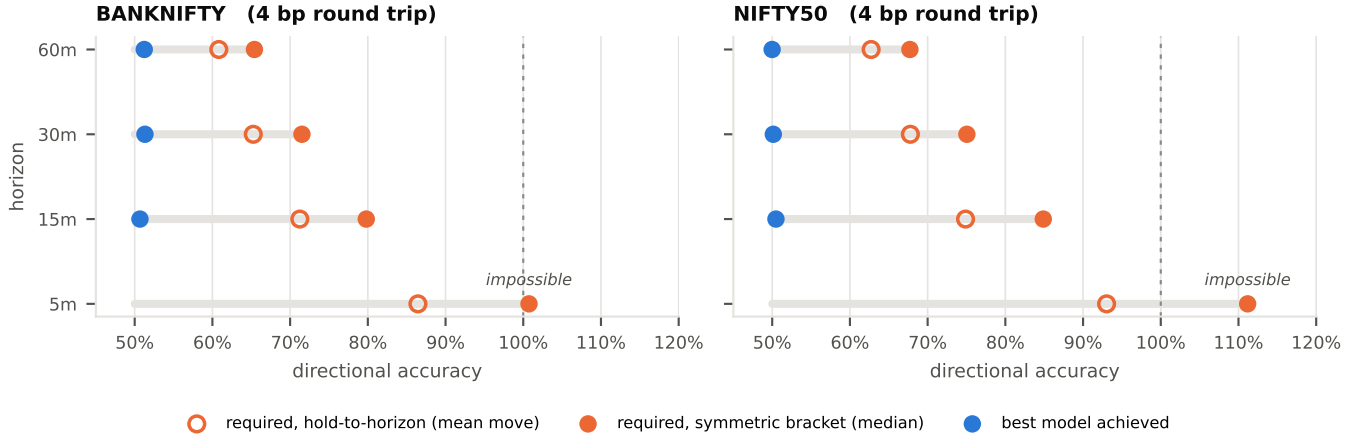


Fig. 4. Break-even accuracy against the best accuracy any model family reached, by horizon. Points beyond the dashed line at 100 % are horizons at which no directional accuracy is sufficient.

TABLE XII
BREAK-EVEN ACCURACY BY COST REGIME, BANKNIFTY
30-MINUTE HORIZON (MEDIAN MOVE 9.3 BP).

Cost	Instrument	p^*
1 bp	Exchange fees only, no slippage	55.4 %
2 bp	Tight futures, minimal slippage	60.8 %
4 bp	Realistic futures round trip	71.5 %
8 bp	Futures with adverse slippage	93.0 %
20 bp	ATM options: spread + 30 min theta	impossible
40 bp	OTM options or a wide book	impossible

100 % and no accuracy suffices. That conclusion is specific to the bracket. Intraday move distributions are strongly right-skewed (skew 2.6–2.9 at five minutes), so the mean absolute move is considerably larger than the median, and a hold-to-horizon rule with no bracket requires 86 % on BANKNIFTY and 93 % on NIFTY 50 instead. Those are not impossible, merely far beyond anything in this paper or in the literature. We report both because quoting only the median would overstate the case, and the point survives either way: the shorter the horizon, the more certainly it fails, which inverts the usual marketing of high-frequency retail strategies.

Instrument choice can make the requirement unreachable. Table XII evaluates (3) across cost regimes.

“Impossible” denotes $c > m$, hence $p^* > 1$: the round trip exceeds the typical move and a perfect forecaster loses on the median trade. The instrument most retail participants actually use, over the horizon they actually hold, has no directional accuracy that makes it profitable.

A. Statistical significance is not economic significance

Section VII reported an apparent BANKNIFTY effect at $p = 0.001$ and then removed it under correct scoring. The effect was economically irrelevant even while it stood: 51.3 % against a 71.5 % requirement loses on every trade,

so the correction changed the statistical verdict without changing the practical one.

That is the more general point. The gap between achievable and required accuracy is roughly twenty percentage points, which is far larger than any effect the scoring dispute could move. Fourteen months of accuracy optimisation produced no tradeable result, and would not have produced one even if the apparent edge had been real.

B. Reframing high-accuracy claims

Suppose a claimed 85 % accuracy at the 30-minute horizon were genuine. It would clear break-even by 13.5 points on BANKNIFTY futures and 9.9 on NIFTY 50, and would remain unprofitable on options. A claim of 85 % is therefore either an artefact of the kind measured in Section V, where a zero-skill system reaches that figure occasionally, or it is true and far less valuable than it sounds. Both readings argue against paying for it.

C. A filter that worked and still lost

A final corroboration. We trained a meta-label filter [18] in which a secondary model decides whether a primary signal’s setup is worth taking, labelled on *points per minute* rather than on target-hit. The label choice matters: training on target-hit raises the hit-rate from 42 % to 53 % while making the money worse, because the model learns to prefer small, easy targets.

Re-run from the committed script on BANKNIFTY, the filter trains on 2085 setups and is tested on 1168 it has never seen. Keeping the best 30 % raises gross quality from +0.78 to +1.29 points per minute of exposure, an improvement of roughly 65 %.

The filter therefore does what it was built to do, on the label it was given. What we cannot show from the committed code is the cumulative cost-aware profit and loss. An earlier version of this paper reported that total moving from −3777 to −981 index points; those figures

TABLE XIII

VOLATILITY PREDICTION THROUGH THE HARNESS OF SECTION IV, 30-MINUTE HORIZON, FIVE PURGED FOLDS. INTERVALS ARE AT THE EFFECTIVE SAMPLE SIZE.

	per bar	per event	CI at n_{eff}
<i>BANKNIFTY</i> ($n_{\text{eff}} = 976$)			
LightGBM	70.8	71.3	[68.0, 73.7]
Persistence baseline	67.4	68.0	[64.5, 70.4]
Shuffled-label control	54.1	—	—
<i>NIFTY 50</i> ($n_{\text{eff}} = 910$)			
LightGBM	72.0	72.1	[69.1, 74.9]
Persistence baseline	67.4	68.5	[64.4, 70.5]
Shuffled-label control	52.2	—	—

do not reproduce and are withdrawn. The verified claim is narrower and still worth making: selecting the best third of setups substantially improves per-minute quality on a primary signal that Section VII shows has no directional edge. Concentrating a null does not create one.

IX. WHAT IS PREDICTABLE INSTEAD

The fifteen production features are all transforms of one price series; reshaping price cannot reveal what price does not contain. What follows comes from changing the target or the information source.

These results have been through the harness of Section IV, and what it leaves behind is narrow enough to state precisely: *none of this project’s findings survives as a finding of ours*. The variance risk premium fails its own multiplicity correction and is withdrawn below. Volatility prediction is untouched by the correction that destroyed the directional result — which matters, because an instrument that flattened everything would be useless — but what it demonstrates is volatility clustering, a regularity documented since Engle [22], and its margin over a persistence baseline is not established. We report it as a working confirmation, not as a discovery.

A. Volatility

The target is binary magnitude: will realised volatility over the next 30 minutes exceed the median of the fold’s own training window? The label is thresholded inside each fold, since thresholding on the full sample would leak the test period’s volatility level. Volatility clusters, so the decisive comparison is not against a coinflip but against *persistence*, a rule that predicts BIG whenever the last 30 minutes were above the same threshold (Table XIII).

Three things separate this from the directional case.

It survives per-event scoring. Accuracy moves from 70.8% to 71.3% on BANKNIFTY and 72.0% to 72.1% on NIFTY 50 when re-scored one observation per horizon. Direction fell from 51.3% to 50.4% under the identical operation. The correction is therefore not an instrument that flattens everything it touches; it distinguishes a real effect from an artefact, which is the only reason to trust what it did in Section VII.

TABLE XIV

SHORT ATM STRADDLE HELD TO EXPIRY, NET OF A ROUND-TRIP COST OF 10% OF PREMIUM COLLECTED. DTE IS ENTRY IN TRADING DAYS BEFORE EXPIRY. p IS ONE-SIDED FOR PROFITABILITY; THE INTERVAL IS A 2000-SAMPLE BOOTSTRAP ON THE MEAN, WHICH ASSUMES NO DISTRIBUTIONAL SHAPE, AND SHORT-PREMIUM P&L IS SHARPLY LEFT-SKEWED.

DTE	n	Win	Net	p	95 % CI
<i>NIFTY</i> (weekly expiries, median gap 7 days)					
1	108	62 %	3.2	0.394	[−21, 26]
2	108	62 %	12.1	0.228	[−20, 43]
3	108	61 %	2.8	0.446	[−39, 43]
5	107	48 %	−45.5	0.967	[−93, 3]
<i>BANKNIFTY</i> (monthly, median gap 27 days)					
1	40	68 %	59.3	0.121	[−43, 149]
2	40	65 %	58.1	0.255	[−117, 214]
3	39	64 %	14.1	0.426	[−124, 153]
5	39	49 %	−209.2	0.979	[−408, −15]

Net in index points. **0 of 4** configurations are significantly profitable on either index, before or after Benjamini–Hochberg; every interval contains zero. Standard deviations run 123–624.

The shuffled control separates cleanly. At 52–54 % against 71–72 % on real labels, the gap is roughly twenty points. In the directional experiment the same gap was under two.

But the edge over persistence is not established. The model beats “the next thirty minutes look like the last thirty” by only +3.3 points on BANKNIFTY and +3.7 on NIFTY 50 per event, and at n_{eff} the model and baseline intervals overlap. Most of a 71 % headline is volatility clustering, which has been documented since Engle [22], [23] and needs no model. What the model adds is real but small, and we do not claim it is significant.

The useful statement is therefore narrower than “72 % accuracy”. It is that magnitude carries far more structure than direction on identical features, and that most of that structure is already in the previous half hour.

B. The variance risk premium

Short at-the-money straddles held to expiry, priced on real end-of-day premiums from the F&O archive. The entry point is a choice, so all four entry points we tried are reported rather than the best one (Table XIV).

It does not survive its own multiplicity correction. No configuration is significantly profitable on either index, and every bootstrap interval contains zero. Recentring each configuration’s P&L to a zero mean and resampling gives the distribution of the best of four under a true null: on NIFTY the median null best is 17.8 against an observed best of 12.1 ($P = 0.68$), and on BANKNIFTY 75.1 against 59.3 ($P = 0.62$). As with the strategy sweep of Section VII-C, the winner is worse than what selection alone typically produces.

The two indices were never comparable. NSE moved BANKNIFTY options from weekly to monthly expiry during the sample, which changes the holding period and the decay profile of the trade being tested. Splitting BANKNIFTY on expiry spacing, the weekly sub-sample

TABLE XV

ALL EIGHT OPTION-CHAIN FEATURES SCREENED AGAINST NEXT-DAY OPEN-TO-CLOSE, BANKNIFTY, 2181 DAYS (MAY 2016 – MAY 2026), AFTER THE PRICE COMPONENT IS REGRESSED OUT. H1 AND H2 ARE THE INFORMATION COEFFICIENT IN EACH HALF OF THE SAMPLE. BH IS BENJAMINI–HOCHBERG AT A 5% FALSE-DISCOVERY RATE ACROSS ALL EIGHT.

Feature (residual)	IC	p	hit	H1	H2	BH
Δ put–call ratio, OI	0.059	0.006	51.3	0.052	0.068	yes
Δ p–c ratio, volume	0.028	0.191	52.1	0.050	−0.000	no
Put–call ratio, OI	0.044	0.041	51.9	0.062	0.045	no
Put–call ratio, volume	0.043	0.043	51.9	0.055	0.040	no
Put–call ratio, Δ OI	0.042	0.047	53.1	0.077	0.005	no
Distance, max pain	−0.045	0.036	52.5	−0.034	−0.050	no
Distance, resistance	−0.030	0.164	51.5	−0.003	−0.057	no
Distance, support	−0.049	0.023	52.1	−0.065	−0.026	no

Price-only features on the same days

Returns, 1–20 days	−0.014 to 0.001	48–53	—	—	no
20-day volatility	−0.030	0.160	50.6	—	no

Six of eight clear $p < 0.05$ uncorrected; **one** survives BH, and the same one survives Bonferroni.

returns −67.5 points per trade and the monthly +57.5: the positive headline is entirely the monthly regime, on 21 expiries. NIFTY has no monthly sub-sample to compare against.

The cost assumption is load-bearing. At 5% of premium, NIFTY returns +17.0 and BANKNIFTY +50.9 per trade; at 20%, −25.6 and −59.4. The sign of the result is set by an assumption, not by the data.

We therefore withdraw the claim, made in an earlier version of this work and repeated in our public research record, that premium selling is the one measurable edge this project found. The variance risk premium is real and well documented [25], [26]; what we can show is that our measurement of it does not survive the standard this paper argues for. Mean premium does exceed mean subsequent move in most configurations, and the win rates of 61–68% are genuine, but a 61% win rate on a left-skewed payoff with a standard deviation forty times the mean is not evidence of an edge. It is the shape that makes short-volatility strategies dangerous.

C. Option-chain positioning

The production features are transforms of one price series, so the natural place to look for something else is what participants have actually committed money to. Open interest is free in the end-of-day archive, and a decade of it is on disk.

The first screen on 350 days found distance-from-max-pain predicting next-day open-to-close at roughly 60%. It also found that feature correlated 0.818 with the plain 10-day return, and that the 10-day return alone did the job slightly better: an apparent options edge that was price mean-reversion wearing an options costume. The corrected procedure is to regress out the price component and test the residual, which is what Table XV reports over 2181 trading days.

Six of the eight residual features clear $p < 0.05$ uncorrected, on days where nothing in the price-only set clears it at all. Applying Benjamini–Hochberg across the eight leaves **one**: the daily *change* in put–call open-interest ratio, which is also the only feature that strengthens in the second half of the decade. Flow beats position, which is consistent with the rest of this study — what participants did today carries more than what they are holding.

An information coefficient of 0.059 over 2181 days is $t \approx 2.8$. That is a real effect and a small one, corresponding to about a 51% hit-rate, which by (3) is nowhere near tradeable on its own.

Two notes on reading the table. The ordering by $|\text{IC}|$ and by hit-rate disagree, which is expected rather than an error: an information coefficient correlates a feature with the *signed* return while a hit-rate counts sign agreement only, so the two diverge whenever a feature carries information about magnitude as well as direction. And an information coefficient is not a trading result; converting one into a strategy would reintroduce every cost question of Section VIII.

D. Edges decay

The strongest next-day signal in an end-of-day options screen is a contrarian rule on put–call ratio extremes. Rerun from the committed script over 489 days of NIFTY data it scores 58% on the 196 days that trigger it, and splits 62% across the first half of the sample against 53% across the second. The same rule on BANKNIFTY scores 49% and is not stable at all.

A sharper decay profile appeared in our research log, quoting 74% in 2024 against 36% and a profit factor of 0.74 in the final six months. Those figures come from a robustness script that was not retained, they do not reproduce from what is committed, and they are withdrawn. The half-sample split above is what we can show, and the finding survives it: a rule that looks strong over one half of a sample can be nine points weaker over the next. This is the mechanism Timmermann and Granger describe [21], whereby discovered patterns self-destruct, and it bears directly on sustained-accuracy claims. If an apparent edge can lose that much between halves of a two-year sample, a claim of stable accuracy over 76 trading days carries little information even when every number in it is true.

X. LIMITATIONS

We state these plainly because the paper’s argument is about honest reporting.

On the estimator. The design effect over-states the dependence by close to a factor of two, and the calibrated variant divides by a constant we fixed at 2.0 rather than deriving. That constant is validated only by coverage, on one data-generating process, at three skill levels. Its input is also imprecisely measured: a single session’s n_{eff} lands between $0.65\times$ and $1.67\times$ the pooled value nine times in

ten, which is wide enough to flip a borderline session’s verdict, as Section VI-C shows on a real one.

On coverage. We evaluate coverage against a synthetic forecaster whose calls persist and whose outcomes come from real bars. That process is calibrated to our production log, but it is one process. A forecaster whose persistence structure differs would need its own check.

On scope. Two indices carry the primary intraday analysis, over seventeen months, in one market. The ceiling test replicates across five, but the model zoo and the strategy sweep do not. Table VI rests on the two best sessions in a log holding five, and the session that motivated the whole project cannot be audited because that portion of the log was lost.

On the cost model. Break-even is computed for a symmetric bracket at the median move and for a hold-to-horizon rule at the mean; both are reported, and they differ by 15 points at the 5-minute horizon. Neither models the path, so the probability of touching a stop before a target is not accounted for.

On what was not re-run. Some figures in our research record come from scripts that were not retained; they are labelled as such in the repository and none carry a claim in Sections V–VIII. Two that were re-run — the meta-label filter’s cumulative P&L and the decay profile of the put–call signal — did not reproduce and are withdrawn in place.

On the data. Index spot carries no volume, so any VWAP-derived feature is misnamed, and whether a true volume-weighted average helps is untested. The Temporal Fusion Transformer is a compact variant retaining variable selection, gated residual networks and interpretable attention while omitting static covariates and the quantile head, and should not be read as the published architecture [33]. Finally, the order-flow hypothesis, where the literature locates genuine short-horizon alpha [27], [28], remains untested here for want of history: we run a live collector, but message-level history is not sold to retail accounts.

XI. CONCLUSION

The headline metric of intraday directional forecasting is not merely noisy. Its *stated precision* is systematically wrong, and wrong by enough to make a coinflip look like a discovery. A forecaster with zero skill posts a $\geq 71\%$ session once in 50, and once every seven sessions somewhere on a modest panel, while reporting a confidence interval $1.66\times$ too narrow at 5-minute granularity and $3.11\times$ too narrow at the per-minute density these systems actually publish. The reading itself is not inflated. The certainty attached to it is.

Applied to our own work, the correction leaves nothing standing. Of nine model families on two indices, four cleared $p < 0.05$ under per-bar scoring and *none* clears it per event or at an effective sample size. Eighty-six rule-based strategies produce a winner worse than a block-bootstrap null. Short premium selling, which we previously

published as the one measurable edge this project had found, fails the same multiplicity correction. The one result that survives untouched — volatility prediction at 71–72% — beats a persistence baseline by three points with overlapping intervals, which is a confirmation of volatility clustering rather than a finding of ours.

So the honest summary is that this project found nothing tradeable, and that the value of the work is in how thoroughly it established that. What we can offer is a method: an effective sample size for overlapping persistent forecasts, a calibrated interval built on it, and a required-versus-achieved framing that turns an accuracy figure into a question about cost. Break-even at the 30-minute horizon is 71.5%, the best per-event accuracy we recorded is 51.2%, and on options over short horizons the requirement rises past the point where any accuracy would do.

Reporting practice should change to match, and the change is cheap. Section VI gives an estimator that covers at its nominal rate where the conventional one covers at 75%, at no cost to the point estimate. Scoring unit, effective sample size, multiple-testing correction, a shuffled-label control and an explicit cost model are not optional refinements to an accuracy figure. Without them the figure does not mean anything, and we know that because we published several that did not.

REPRODUCIBILITY

Code, the harness, raw results, figure generation and the full research record, including every retracted claim, are at <https://github.com/rachitranka25/accuracy-illusion>. Experiments use a fixed seed and single-threaded native pools.

REFERENCES

- [1] L. P. Hansen and R. J. Hodrick, “Forward exchange rates as optimal predictors of future spot rates: An econometric analysis,” *Journal of Political Economy*, vol. 88, no. 5, pp. 829–853, 1980.
- [2] M. Richardson and T. Smith, “Tests of financial models in the presence of overlapping observations,” *The Review of Financial Studies*, vol. 4, no. 2, pp. 227–254, 1991.
- [3] J. Boudoukh, M. Richardson, and R. F. Whitelaw, “The myth of long-horizon predictability,” *The Review of Financial Studies*, vol. 21, no. 4, pp. 1577–1605, 2008.
- [4] B. R. Moulton, “An illustration of a pitfall in estimating the effects of aggregate variables on micro units,” *The Review of Economics and Statistics*, vol. 72, no. 2, pp. 334–338, 1990.
- [5] L. Kish, *Survey Sampling*. New York, NY, USA: Wiley, 1965.
- [6] W. K. Newey and K. D. West, “A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix,” *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.
- [7] H. R. Künsch, “The jackknife and the bootstrap for general stationary observations,” *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.
- [8] D. N. Politis and J. P. Romano, “A circular block-resampling procedure for stationary data,” in *Exploring the Limits of Bootstrap*, R. LePage and L. Billard, Eds. New York, NY, USA: Wiley, 1992, pp. 263–270.
- [9] D. N. Politis and J. P. Romano, “The stationary bootstrap,” *Journal of the American Statistical Association*, vol. 89, no. 428, pp. 1303–1313, 1994.
- [10] M. H. Pesaran and A. Timmermann, “A simple nonparametric test of predictive performance,” *Journal of Business & Economic Statistics*, vol. 10, no. 4, 1992.
- [11] Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate: A practical and powerful approach to multiple testing,” *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.

- [12] H. White, "A reality check for data snooping," *Econometrica*, vol. 68, no. 5, pp. 1097–1126, 2000.
- [13] R. Sullivan, A. Timmermann, and H. White, "Data-snooping, technical trading rule performance, and the bootstrap," *The Journal of Finance*, vol. 54, no. 5, pp. 1647–1691, 1999.
- [14] P. R. Hansen, "A test for superior predictive ability," *Journal of Business & Economic Statistics*, vol. 23, no. 4, pp. 365–380, 2005.
- [15] C. R. Harvey, Y. Liu, and H. Zhu, "...and the cross-section of expected returns," *The Review of Financial Studies*, vol. 29, no. 1, pp. 5–68, 2016.
- [16] D. H. Bailey and M. López de Prado, "The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting and non-normality," *The Journal of Portfolio Management*, vol. 40, no. 5, pp. 94–107, 2014.
- [17] D. H. Bailey, J. Borwein, M. López de Prado, and Q. J. Zhu, "The probability of backtest overfitting," *Journal of Computational Finance*, vol. 20, no. 4, pp. 39–70, 2017.
- [18] M. López de Prado, *Advances in Financial Machine Learning*. Hoboken, NJ, USA: Wiley, 2018.
- [19] E. F. Fama, "Efficient capital markets: A review of theory and empirical work," *The Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.
- [20] A. W. Lo and A. C. MacKinlay, "Stock market prices do not follow random walks: Evidence from a simple specification test," *The Review of Financial Studies*, vol. 1, no. 1, pp. 41–66, 1988.
- [21] A. Timmermann and C. W. J. Granger, "Efficient market hypothesis and forecasting," *International Journal of Forecasting*, vol. 20, no. 1, pp. 15–27, 2004.
- [22] R. F. Engle, "Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation," *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.
- [23] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [24] A. R. Admati and P. Pfleiderer, "A theory of intraday patterns: Volume and price variability," *The Review of Financial Studies*, vol. 1, no. 1, pp. 3–40, 1988.
- [25] P. Carr and L. Wu, "Variance risk premiums," *The Review of Financial Studies*, vol. 22, no. 3, pp. 1311–1341, 2009.
- [26] G. Bakshi and N. Kapadia, "Delta-hedged gains and the negative market volatility risk premium," *The Review of Financial Studies*, vol. 16, no. 2, pp. 527–566, 2003.
- [27] P. N. Kolm, J. Turiel, and N. Westray, "Deep order flow imbalance: Extracting alpha at multiple horizons from the limit order book," *Mathematical Finance*, vol. 33, no. 4, pp. 1044–1081, 2023.
- [28] Z. Zhang, S. Zohren, and S. Roberts, "DeepLOB: Deep convolutional neural networks for limit order books," *IEEE Transactions on Signal Processing*, vol. 67, no. 11, pp. 3001–3012, 2019.
- [29] S. Gu, B. Kelly, and D. Xiu, "Empirical asset pricing via machine learning," *The Review of Financial Studies*, vol. 33, no. 5, pp. 2223–2273, 2020.
- [30] T. Fischer and C. Krauss, "Deep learning with long short-term memory networks for financial market predictions," *European Journal of Operational Research*, vol. 270, no. 2, pp. 654–669, 2018.
- [31] C. Krauss, X. A. Do, and N. Huck, "Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500," *European Journal of Operational Research*, vol. 259, no. 2, pp. 689–702, 2017.
- [32] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu, "Financial time series forecasting with deep learning: A systematic literature review: 2005–2019," *Applied Soft Computing*, vol. 90, art. 106181, 2020.
- [33] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.
- [34] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [35] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.
- [36] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [37] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 3146–3154.
- [38] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [39] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.